

DEFINITION — HOW DATA IS DESCRIBED

API INTERFACES6

**ODCS**
BITOL @ LF

**OpenAPI**
LF

**AsyncAPI**
LF

**GraphQL**
LF

**gRPC**
CNCF

**OData**
OASIS

DATA PRODUCTS4

**ODPS**
BITOL @ LF

**DPDS**
ODM

**ODPSpec**
LF

**DPROD**
OMG

SCHEMA5

**XML Schema**
W3C

**JSON Schema**
IETF

**SQL DDL**
ISO/IEC

**AVRO Schema**
ASF

**Protobuf**
GOOGLE

SEMANTICS6

**OSI**
OSI INITIATIVE

**RDF/OWL**
W3C

**DCAT**
W3C

**SKOS**
W3C

**SHACL**
W3C

**JSON-LD**
W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS6

**CSV**
IETF

**JSON**
IETF

**XML**
W3C

**PARQUET**
ASF

**AVRO**
ASF

**ORC**
ASF

OPEN TABLE FORMATS3

**Iceberg**
ASF

**Delta**
LF

**Hudi**
ASF

STORAGE SYSTEMS2

**S3**
AWS

**HDFS**
ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY3

**JDBC**
JCP

**ODBC**
ISO/IEC

**ADBC**
ASF

INTERCONNECTION3

**HTTP**
IETF

**FTP / SFTP**
IETF

**Delta Sharing**
LF

MESSAGING5

**Kafka**
ASF

**MQTT**
OASIS

**CloudEvents**
CNCF

**AMQP**
OASIS

**JMS**
JAKARTA EE

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2

**DataFrame API**
DATA-APIS

**Apache Arrow**
ASF

PROCESSING4

**dbt**
DBT LABS

**Spark**
ASF

**Pandas**
NUMFOCUS

**Ibis**
IBIS

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS4

**Iceberg Catalog**
ASF

**Unity Catalog**
LF

**Hive Metastore**
ASF

**Schema Registry**
CONFLUENT

LINEAGE1

**OpenLineage**
LF

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY2

**SQL**
ISO/IEC

**SPARQL**
W3C

DATA QUALITY3

**dbt**
DBT LABS

**Great Expectations**
GX LABS

**SodaCL**
SODA

OBSERVABILITY2

**OpenTelemetry**
CNCF

**OORS**
BITOL @ LF

POLICIES1

**OPA**
CNCF

AI INTERFACES2

**MCP**
LF

**A2A**
LF

☐ Open standard

☒ Highlighted by Entropy Data — the ones to bet on today

☐ Emerging / draft

☐ Single-vendor — muted